Lecture 1. Data, Similarity Measures, Preprocessing, and Transformation of Data

Data Mining, prof. Biljana Tojtovska Ribarski
FCSE, Skopje

Contents

1 Similarity, Preprocessing, and Transformation in Data Mining

1.1 Introduction

In data mining, one of the most common hidden assumptions is that once data have been encoded as vectors, the problem of comparison has been solved. This is not true. A representation tells us what the objects look like numerically, but it does not by itself tell us what it means for two objects to be *close*, *similar*, or *different*. A correct analysis therefore requires two decisions: how the data should be represented, and how that representation should be compared.

The core message of this chapter is that similarity, distance, preprocessing, and transformation must be chosen in a way that respects the meaning of the data and the question being asked. A transformation that is perfectly reasonable for one task may be wrong for another. A distance that works well for numeric vectors may fail badly for sets, graphs, road networks, sequences, or probability distributions. Even within the same dataset, different goals may justify different comparison rules.

1.2 A Guiding Framework for Choosing a Measure

A useful way to read almost any comparison problem is through the sequence

$$\textit{Data} \rightarrow \textit{Task} \rightarrow \textit{Compare} \rightarrow \textit{Measure} \rightarrow \textit{Why it fits / Pitfall.}$$

Many mistakes arise from skipping one of the middle steps.

Definition: Comparison-aware analysis

A *comparison-aware analysis* is an analysis in which the representation of the data, the comparison target, and the comparison rule are all chosen to match the scientific, business, or algorithmic question of interest.

Remark

Two objects may be represented by the same numeric structure and still require different comparison rules, because the notion of "difference" may change from one task to another. For example, a route can be compared by geometric shape, by travel time, by overlap in road segments, or by turn-by-turn sequence similarity.

1.3 When Are We *Not* Allowed to Transform Data?

Transformation is often useful, but it is not automatically harmless. The first question is not "Can this be standardized?" but rather "What would be changed conceptually if we standardized it?"

1.3.1 When the Transformation Changes the Meaning of the Attribute

Consider daily temperatures for ten cities measured in degrees Celsius, with the question: *Which two cities have similar climate?* The data objects are temperature series, the comparison is city versus city, and the scientific meaning concerns pattern similarity, not ratio claims.

Warning

If one normalizes Celsius values and then interprets the result using ratio language such as "twice as hot," the analyst is treating an interval-scale variable as if it were ratio-scale. Celsius has an arbitrary zero, so ratio statements are not meaningful in the same way they would be for a genuine ratio-scale variable.

Insight

Transformations must respect the measurement scale of the attribute. If the transformation changes the meaning of the scale, it may change the meaning of the answer.

1.3.2 When Regulation or Auditing Requires Original Values to Be Recoverable

Suppose the data consist of hospital billing records containing monetary amounts, timestamps, and codes, and the task is to compare two fraud-detection or anomaly-detection model versions. In such a setting, derived features may be helpful, but the raw values must remain recoverable.

If only standardized or transformed values are retained, the exact original evidence trail may be lost. This is unacceptable in regulated environments because auditability depends on reproducing the original case record.

Remark

The usual professional practice is not "replace raw data by transformed data," but rather "retain raw data and add transformed features as additional fields."

1.3.3 When the Scientific Question Is About Absolute Magnitude

Consider campus energy consumption measured in kilowatt-hours per building per month, with the question: *Which building consumes the most energy?* Here the correct comparison concerns absolute totals.

If one z-scores each building separately, the problem changes. It no longer answers which building consumes the most. Instead, it answers which building has a month that is unusual relative to its own baseline. That may be a useful question, but it is a *different* question.

Warning

A transformation can silently convert an absolute-comparison problem into a relative-within-object problem.

1.3.4 When Variance Is the Signal

Suppose multiple sensors measure vibration amplitude on machines, and the question is: *Which machine is becoming unstable?* In this setting, growing variability may itself be the warning signal.

Standardizing each sensor to unit variance erases the very phenomenon we are trying to detect. A sensor that becomes more variable than before may stop looking unusual once its variance has been normalized away.

Insight

Sometimes the spread, variance, or amplitude structure is the information of interest. In such cases, variance-normalizing transformations destroy signal rather than remove nuisance variation.

1.3.5 Example: Scaling Changes the Geometry and Can Flip Clustering

A simple two-dimensional example makes this concrete. Let Feature 1 contain the real signal, so that two clusters are separated mainly along that axis. Let Feature 2 be high-variance noise. If we run k - means with Euclidean distance on the raw variables, Feature 2 dominates the geometry because it has the larger numerical scale. After standardization, Feature 1 and Feature 2 receive comparable influence, and the clustering can change.

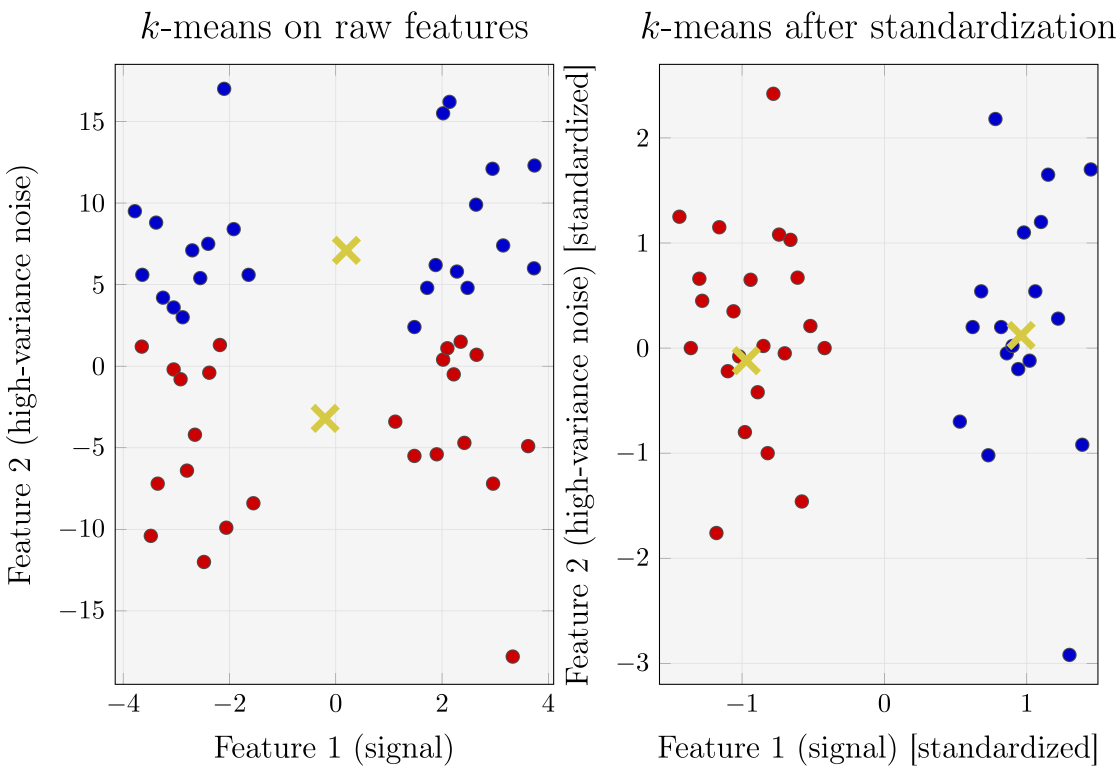

Figure 1: Scaling can flip clustering: k-means on raw features (left) versus after standardization (right).

1.4 Metrics and Non-Metrics: Why Properties Matter?

Not every quantity that is called a "distance" behaves like a metric. This matters because many algorithms, data structures, and geometric intuitions assume metric properties.

1.4.1 Why Metric Properties Matter in Practice

Nearest-neighbor indexing structures (such as VP - trees, GNAT, and other metric trees) exploit the triangle inequality to prune candidates safely. If triangle inequality fails, those pruning rules can eliminate the true nearest neighbor.

Definition: Metric

A function $d(x, y)$ is a metric if it satisfies nonnegativity, symmetry, identity of indiscernibles, and triangle inequality. (For details see the slides.)

1.4.2 A Quantity That Looks Like a Metric but Is Not

A clean example is cosine distance defined by

$$d(x, y) = 1 - \cos(x, y).$$

Take $x = (1, 0)$ and $y = (2, 0)$. Their cosine similarity is 1, so $d(x, y) = 0$, even though $x \neq y$. Thus identity of indiscernibles fails.

Warning

A distance - like quantity can return zero for two nonidentical vectors. This means it is not a true metric, even if it is widely used in practice.

Dynamic Time Warping (DTW) provides a second cautionary example. DTW is often very useful for time series, but it can fail both triangle inequality and identity of indiscernibles. This motivates semi-metric corrections in some settings.

1.4.3 Example Where Cosine Similarity Is Zero

If $x = (1, 0)$ and $y = (0, 1)$, their dot product is zero and therefore cosine similarity is zero. In text mining, the same phenomenon occurs when two bag-of-words vectors have no shared terms. A zero cosine here means "no overlap in direction" under the chosen representation; it does *not* necessarily mean semantic unrelatedness.

1.4.4 Why Use General Minkowski Distances Instead of Euclidean?

The Minkowski family lets us choose a notion of deviation that matches the task. Consider query point $q = (0, 0, 0)$ and two candidates

$$a = (2, 0, 0), \qquad b = (1, 1, 1).$$

Under L_2, candidate b is closer because $\sqrt{3} < 2$. Under L_1, candidate a is closer because $2 < 3$.

The L_2 norm treats many small deviations as accumulating in an energy-like way, while L_1 sums absolute deviations and is often preferred when sparse changes or robustness to one - feature spikes matter.

A quality-control example makes the role of L_∞ clear. Suppose an item is measured by length, width, and thickness, and the task is to reject it if *any one* dimension deviates too much. Then

$$d_\infty(x, y) = \max_j |x_j - y_j|$$

fits the business rule, because the worst dimension is what matters.

1.5 Non-Standard Data Types and the Measures That Match Them

Many important data-mining objects are not best understood as ordinary Euclidean vectors. Even when they can be encoded numerically, the correct comparison measure often comes from their combinatorial or structural meaning rather than from vector geometry alone.

1.5.1 Sets: Jaccard and Dice for Presence/Absence Data

Suppose each customer order is represented as a set of purchased items, as in market-basket data. The task is to find customers with similar purchasing types, not customers who bought the same total quantity.

In this setting, Jaccard or Dice overlap is natural because the question concerns shared assortment, not repeated counts. If repeated quantities should not dominate the comparison, a set-based similarity is preferable to Euclidean distance on counts.

*1.5.2 Strings and Sequences: Edit Distance and Alignment Distance**

DNA sequences, barcodes, and other symbolic strings are compared by asking how many operations are needed to turn one sequence into another. The operational meaning is therefore not geometric displacement in ^{p}, but substitution, insertion, deletion, or alignment cost.

Definition: Edit distance viewpoint

For sequence data, a distance is often defined as the minimum cost of transforming one sequence into another through allowable edit or alignment operations.

In bioinformatics, alignment-based distances such as those derived from Needleman–Wunsch, are canonical because biological similarity is often better captured by aligned correspondence than by coordinate - wise comparison.

1.5.3 *Example: Mixed Data and the Failure of One-Hot Euclidean Distance*

Consider a small mixed-type table of six customer or person records. Some variables are numeric (age, income), some ordinal (education level), some categorical (occupation, city), and one is asymmetric binary (a rare symptom where presence matters more than joint absence). The task is nearest-neighbor search or clustering of similar people.

A common but problematic strategy is to one-hot encode the categorical variables, min–max scale the numeric ones, and then apply Euclidean distance. Gower-style distance is specifically designed for mixed-type data. It normalizes numeric variables by range, handles categorical mismatch as 0/1, normalizes ordinal levels, and can treat asymmetric binary variables in a presence-focused way.

Warning

One-hot encoding is not "wrong" by itself, but Euclidean distance on one-hot vectors can impose a geometry that does not match the mixed-type semantics of the task.

1.5.4 *Time Series: Alignment-Based Distance**

Time series from the UCR archive provide a standard example. If the same event occurs at slightly different times or speeds in two series, Euclidean pointwise comparison may declare them far apart even when they have the same underlying shape.

DTW addresses this by aligning the series nonlinearly before measuring mismatch. The price of that flexibility is that DTW may fail metric axioms, which matters for indexing assumptions and some algorithmic guarantees.

1.5.5 *Distributions as Data Objects**

Sometimes the observation itself is a probability distribution or histogram. In that case, the comparison must reflect how mass moves between bins, not merely how bin counts differ numerically.

Example: Image retrieval

Represent each image by a color histogram. The task is to retrieve images that are visually similar to a query image. If color mass shifts slightly from dark blue to light blue, bin-by-bin Euclidean distance can exaggerate the difference. Earth Mover's Distance (EMD), or more broadly the Wasserstein idea, measures the minimum work needed to move mass from one histogram to another and therefore matches the visual intuition more closely.

Example: Before versus after policy comparison

Represent commute times before and after a transportation change as two distributions. The task is to ask whether the new system changed the shape of commute times, perhaps by shrinking the tail while leaving the mean similar. Wasserstein distances are appropriate because they respect the geometry of where probability mass moves. In contrast, KL divergence is not a metric, is asymmetric, and can blow up when one distribution assigns zero probability where the other does not.

1.6 Graphs, Trees, and Structural Comparison

Graphs and trees are structurally rich objects. Their comparison is rarely meaningful if reduced to naive Euclidean comparison of adjacency matrices or node-order-dependent encodings.

1.6.1 *Graphs and Networks*

Ego - network data such as SNAP ego - Facebook graphs are a good example. A common task is social - circle discovery or social - structure comparison. The question is not how different two adjacency matrices are numerically under an arbitrary node ordering, but how much structural editing is required to turn one pattern into another.

Two concrete comparison strategies are particularly instructive:

- **Graph Edit Distance (GED):** the minimal edit cost required to transform one graph into another. This is useful for nearest - neighbor retrieval of structurally similar social patterns.
- **Weisfeiler-Lehman (WL) graph kernel:** a similarity based on matching subtree - like patterns, often used with kernel classifiers such as SVMs.

Warning

Adjacency matrix Euclidean distance depends on node ordering. Unless the nodes are already aligned meaningfully, the comparison is often meaningless.

1.6.2 *Trees and Hierarchies*

WordNet-style hierarchies provide a canonical example of tree-structured data. If a predicted taxonomy is compared with a reference taxonomy, the natural question is how many insertions, deletions, and relabelings are needed to transform one hierarchy into the other.

Tree Edit Distance (TED) gives a direct operational answer to that question. This is a structural error measure, not merely a node - level accuracy measure. From a geometric point of view, hierarchical data are also well known to be more naturally embedded in hyperbolic rather than Euclidean spaces.

1.7 Data Needing Non-Euclidean Geometries

Some domains are naturally curved, constrained, or graph-structured. In such settings, forcing Euclidean geometry can produce misleading results even before any algorithm is run.

1.7.1 *Spherical Geometry: GPS and GeoLife Trajectories*

Suppose each record is a route represented as a time-ordered sequence of latitude-longitude points. The task is route retrieval or clustering of similar trips.

The point - level comparison should be geodesic (great-circle) distance on the Earth's surface. For whole trajectories, one may then build curve or alignment distances such as Fréchet or DTW using geodesic point-to-point distances internally.

The reason is straightforward. Latitude and longitude are angular coordinates, not Cartesian coordinates. One degree of longitude represents different physical distances at different latitudes. Moreover, the dateline makes naive coordinate subtraction particularly misleading.

Example: Dateline wrap-around

Let $A = (0°, 179°)$ and $B = (0°, -179°)$. Naively, the longitude difference appears to be $358°$, which looks enormous. But because longitudes wrap around the globe, the effective difference is

$$\Delta\text{lon}_{\text{min}} = \min(358°, 360° - 358°) = 2°.$$

These points are therefore close on Earth even though they look far apart numerically.

1.7.2 When Spherical Geometry Is Still Not Enough: Street-Network Distance

Geodesic closeness is still not the right concept when movement is constrained by a road network. If two points are separated by a river, they may be geographically close and yet require a long detour by road.

In a road graph, intersections are nodes and road segments are weighted edges. The relevant comparison is shortest-path cost,

$$d(u,v) = \min_{\text{paths } u\to v} \sum \text{edge weights.}$$

This can represent travel time, driving distance, walking cost, toll burden, or other operational notions of accessibility.

Example: Across the river failure of spherical distance

Two points are only 200 meters apart in a straight line across a river. Geodesic distance suggests they are very close. But if the nearest bridge is 1.5 km upstream, the shortest driving route may be 3.2 km and about 6 minutes. For the question "Which hospital is closest by driving time?" the geodesic answer is wrong because it answers the wrong problem.

Example: Graph

Let nodes A (user), B (bridge entrance), and C (destination) be connected by one-way edges $A \to B$ with weight 2 minutes and $B \to C$ with weight 4 minutes, with no direct road from A to C. Then

$$d(A,C) = 2 + 4 = 6 \text{ minutes.}$$

On a directed road network, one may also have $d(C,A) \neq d(A,C)$, so the result can be a quasi - metric rather than a symmetric metric.

Remark: Similarity versus cost on roads

Road maps support two fundamentally different forms of comparison. Similarity-based comparison treats a route or road as an object and asks whether two routes look alike or share structure. Cost-based comparison assigns travel costs to road segments and asks for the route that minimizes total cost. Navigation and dispatch systems use the second.

1.7.3 Trajectory Similarity on Street Networks

If the object is a full trip rather than a pair of points, then multiple comparison notions become reasonable. Suppose a map - matched trip is a sequence of road segments,

$$T_1 = (e_5, e_{12}, e_{12}, e_7, e_9), \qquad T_2 = (e_5, e_{12}, e_7, e_{10}).$$

One may compare such routes in at least three ways:

1. **Cost-based similarity:** compare aligned positions by shortest - path travel cost and ask how far apart the trips are in time or road distance.

2. **Path-overlap similarity:** convert each trip to a set of road segments and compute Jaccard overlap,
$$J(T_1,T_2) = \frac{|E(T_1) \cap E(T_2)|}{|E(T_1) \cup E(T_2)|}.$$

3. **Sequence similarity:** treat the edge sequence as a string and use edit distance to count inserted, deleted, or substituted segments.

Example: Route comparison

Let

$$T_1 = (a, b, c, d, e), \qquad T_2 = (a, b, c, f, e).$$

Then $E(T_1) = \{a, b, c, d, e\}$ and $E(T_2) = \{a, b, c, f, e\}$, so

$$J(T_1, T_2) = \frac{|\{a, b, c, e\}|}{|\{a, b, c, d, e, f\}|} = \frac{4}{6}.$$

An edit-distance view would say that the routes differ by one substitution, namely $d \to f$, suggesting a small detour.

Insight

Even on the same street-network data, the chosen similarity must match the question: shared roads, turn-by-turn changes, or actual travel cost are not the same concept.

1.7.4 Compositional Data

Consider yearly budget shares such as

$$\text{Year 1} = (0.50, 0.30, 0.20), \qquad \text{Year 2} = (0.60, 0.25, 0.15),$$

where the components sum to 1. The task is to decide whether the allocation changed meaningfully.

Euclidean distance on the raw shares is often misleading because the sum-to-one constraint induces dependence: increasing one component necessarily decreases others. In compositional data, the natural question is often about ratios between parts rather than raw coordinate differences. Aitchison distance, which is Euclidean distance after a centered log-ratio transformation, is designed for this setting.

1.7.5 Hyperbolic Geometry

The hyperbolic case has already appeared implicitly in the discussion of trees and taxonomies. Hierarchical structure is often represented more faithfully in hyperbolic geometry than in Euclidean geometry, because exponential expansion away from the origin is more compatible with tree - like growth.

1.8 Correlations Beyond Pearson, Spearman, and Kendall

Correlation is another domain in which one measure rarely fits all tasks.

1.8.1 Distance Correlation

If $Y = \sin(X) + \text{noise}$, Pearson correlation may be close to zero even though the variables are dependent. Distance correlation is designed to detect general dependence and equals zero only under independence.

1.8.2 MIC for Exploratory Discovery

When scanning thousands of variable pairs in a biomedical table, one may wish to flag a wide range of interesting relationship shapes. The maximal information coefficient (MIC) is useful for exploratory discovery, although it should be followed by confirmatory analysis because discovery-oriented measures trade off power and false positives in different ways.

1.8.3 Mixed-Type Associations

For mixed data types, specialized association measures are often more appropriate than forcing everything into a single generic correlation coefficient:

- **Point-biserial correlation:** continuous versus binary variables, for example exam score versus pass/fail.

- **Phi coefficient:** binary versus binary variables, for example smoker yes/no versus disease yes/no.

1.8.4 Circular Correlation

Wind direction illustrates why ordinary linear correlation can fail for angles. Directions 359° and 1° are close on the circle but look far apart under naive linear subtraction. Circular or directional statistics are therefore needed.

1.8.5 Cross-Correlation

When marketing spend and sales are both time series, the interesting question may be whether spend leads sales by a lag such as two weeks. Cross-correlation across lags is designed for this lead-lag structure.

1.9 How a Wrong Metric Breaks an Algorithm?

Metric choice is not only interpretive; it directly affects algorithmic behavior.

1.9.1 k-NN and Clustering Dominated by Scale

If customer records contain age on a 0- 100 scale and income on a 0-100000 scale, Euclidean distance without scaling will mostly compare incomes. Neighbors then become "people with similar income," while age contributes almost nothing.

1.9.2 Mixed-Type Data and Euclidean Failure

The UCI Adult dataset mixes categorical and numeric variables. One-hot Euclidean distance can create strange geometry in which rare categories receive disproportionate influence. A better approach is often to compute Gower distance and then use a clustering method that accepts a dissimilarity matrix, such as k-medoids or hierarchical clustering.

1.9.3 Time Series Classification Failure

If time-series patterns occur at variable speed, Euclidean pointwise distance can fail because it penalizes temporal misalignment. An alignment-based measure such as DTW is often better, provided we remember its non - metric caveats.

1.10 Do Algorithms Assume Metric Properties?

Different algorithms rely on different kinds of comparison structure.

- **Euclidean - specific methods:** classical k - means uses centroids and is inherently Euclidean in its standard form.
- **Metric - assumption methods:** metric trees use triangle inequality for pruning.
- **More general dissimilarity - based methods:** some methods, such as k-medoids and certain hierarchical clustering procedures, can operate on arbitrary dissimilarity matrices, though one may lose geometric guarantees and speedups.

Remark

The question is therefore not "Does the algorithm use distance?" but rather "What structural assumptions does the algorithm make about that distance?"

1.11 Data Mining Beyond Prediction

Data mining is broader than supervised prediction. Important goals include:

- **Process mining:** discover actual workflows, detect bottlenecks, and perform conformance checking, for example on event logs such as the BPI Challenge 2012 dataset.

- **Change-point or drift detection:** detect when the generating process changes over time.
- **Causal discovery:** infer plausible causal graph structure rather than only statistical association.
- **Anomaly explanation:** explain why a record is anomalous, not merely flag it.

1.12 One Dataset, Many Metrics for Different Questions

GeoLife trajectories provide a particularly clear example of why one dataset can support several valid comparison measures.

1. **Route-shape similarity:** use Fréchet distance between curves when the question is whether the same map path was followed.
2. **Coverage similarity:** use Hausdorff distance on point sets when the question is whether two trajectories pass through the same places irrespective of order.
3. **Behavior similarity:** compute a speed or acceleration time series and compare it using alignment distances when the question concerns travel behavior, such as walking versus bus riding.

Insight

The same raw data can support different scientifically meaningful distances because the task changes what it means to be similar.

1.13 Distances for Earth and Atmosphere Objects: Problem $\rightarrow$ Metric

Domain meaning determines the appropriate distance.

1.13.1 How Far Are Two Cities on Earth?

The relevant quantity is shortest path on the Earth's surface, so great-circle or geodesic distance is appropriate. Euclidean distance on latitude and longitude distorts physical distance, especially near the poles and the dateline.

1.13.2 Which Hospital Is Closest by Driving?

"Closest" means shortest driving time subject to road constraints, not shortest straight - line distance. The appropriate measure is shortest-path travel time on a road graph, computed for example by Dijkstra's algorithm or A*.

1.13.3 Which Areas Can Reach a School Within Fifteen Minutes?

This is an accessibility or catchment problem. The relevant quantity is minimum travel time $d_T(s, v)$ from the school to each location v. The fifteen - minute isochrone is the set of all points satisfying $d_T(s, v) \leq T$. Such regions are not circles; network structure creates arms and holes.

1.13.4 Are Two Commuting Routes the Same?

This depends on what "the same" means. Shared roads suggest Jaccard overlap on edge sets. Turn-by-turn similarity suggests edit distance on segment sequences. Comparable travel cost suggests cost-based alignment using shortest-path distances between aligned points.

1.13.5 Which Flight Route Is Shortest?

Airline planning is driven by time, fuel, restricted airspace, altitude, and safety, not only by geometric length. A cost function over permitted corridors is therefore more relevant than simple great-circle distance.

1.13.6 Which City Will Be Affected First by Smoke or Pollution?

Atmospheric spread follows flow rather than straight lines. A flow-based or Lagrangian travel-time notion is therefore more appropriate. A city farther away geographically but downwind may be effectively "closer" than a nearby city that is upwind.

1.14 When to Use Similarity Versus Distance

Similarity and distance are related, but they do not always play the same role in modeling and algorithms.

Definition: Similarity-oriented problem

A problem is *similarity-oriented* when the natural output is "higher means more alike," as in ranking, retrieval, or similarity graph construction.

Definition: Distance-oriented problem

A problem is *distance-oriented* when the natural output is "lower means more alike," especially when neighborhoods, metric balls, or minimization procedures are central.

Similarity is often more interpretable for retrieval and ranking tasks. It is also natural when building a weighted similarity graph or when the downstream method explicitly uses a kernel or similarity.

Distance is usually the more natural interface for clustering, nearest - neighbor search, density - based neighborhoods, and embedding procedures. Standard examples include k-means, hierarchical clustering, DBSCAN, OPTICS, MDS, t-SNE, and UMAP.

When a similarity s is bounded in $[0, 1]$, one common conversion is

$$d = 1 - s.$$

For Jaccard similarity this yields a proper metric distance. For cosine similarity, however, $1-\cos(\cdot)$ is commonly used but is not always a true metric; angular distance is sometimes preferable.

Insight

A useful rule of thumb is this: if the goal is ranking, similarity is often the more interpretable language; if the goal is clustering, neighborhoods, or embedding, distance is usually the more natural interface.

1.15 Exercise

Example: Mixed binary similarity with asymmetric and symmetric variables

Consider four binary patient variables:

- x_1: cough present (asymmetric; only 1 matters),
- x_2: fever present (asymmetric; only 1 matters),
- x_3: vaccinated (symmetric),
- x_4: smoker (symmetric).

Two patients are encoded as

$$x = [1, 0, 1, 0], \qquad y = [0, 0, 1, 1].$$

Let $A = \{1, 2\}$ be the asymmetric variables and $S = \{3, 4\}$ the symmetric variables.
Task.

1. Compute the asymmetric counts on A:

$$a = \#(1,1), \qquad b = \#(1,0), \qquad c = \#(0,1).$$

Then compute Jaccard similarity on A:

$$J_A = \frac{a}{a+b+c}.$$

2. Compute the number of matches on the symmetric variables:

$$\text{matches}_S = \#\{i \in S : x_i = y_i\}, \qquad m = |S|.$$

Then compute simple matching on S:

$$\text{SMC}_S = \frac{\text{matches}_S}{m}.$$

3. Compute the combined mixed - binary similarity and distance:

$$\text{sim}(x,y) = \frac{a + \text{matches}_S}{(a+b+c)+m}, \qquad d(x,y) = 1 - \text{sim}(x,y).$$

Answer. On A, we have $a = 0$, $b = 1$, and $c = 0$, so $J_A = 0$. On S, we have $\text{matches}_S = 1$ and $m = 2$, so $\text{SMC}_S = 1/2$. Therefore,

$$\text{sim}(x,y) = \frac{0+1}{1+2} = \frac{1}{3}, \qquad d(x,y) = 1 - \frac{1}{3} = \frac{2}{3}.$$

1.16 Practical Do's and Don'ts

Do's

- Start with the question, not with the formula.
- Check whether the representation respects the scale and meaning of the variables.
- Use transformations that remove nuisance variation only when they do not erase signal.
- Match the measure to the data type: sets, sequences, graphs, routes, and distributions often need specialized comparisons.
- Ask whether the algorithm requires Euclidean or metric structure.
- Use multiple metrics when the problem has multiple risk dimensions.
- Be explicit about the invariances your task assumes, such as translation invariance for images or timing invariance for time series.

Don'ts

- Do not standardize automatically without asking what question the transformed data now answer.
- Do not assume cosine similarity is semantic by itself.
- Do not use Euclidean distance on latitude and longitude as if they were Cartesian coordinates.

- Do not compare mixed-type records with one-hot Euclidean distance without checking whether the induced geometry makes sense.
- Do not assume that two routes, two graphs, or two distributions should be compared by ordinary vector distance simply because they can be numerically encoded.
- Do not optimize only one benchmark or one metric and then infer overall quality.

1.17 Summary

Similarity, distance, preprocessing, and transformation are not merely technical details added after modeling. They are part of the modeling decision itself. A correct distance is one that reflects the scientific, operational, or algorithmic meaning of difference for the task at hand.

A central theme of this chapter is that transformations must preserve meaning. Standardization can help when a feature scale is nuisance variation, but it can be wrong when absolute magnitude matters, when raw values must remain auditable, or when variance is itself the signal. Likewise, Euclidean distance is useful in many familiar numeric settings, but it can fail on mixed data, translated images, aligned sequences, routes on road networks, compositional data, and distributions.

The same dataset can support multiple valid comparison rules. Thus, before computing any similarity or distance, first decide what the comparison is supposed to mean.

Questions

1. Explain the difference between nominal, ordinal, interval, and ratio attributes. For each type, state which comparisons or operations are meaningful and give one example.
2. What is meant by data quality in data mining? List and briefly explain several common data quality problems, such as noise, outliers, missing values, wrong data, fake data, and duplicate data.
3. Explain the difference between noise and an outlier. Why is it not always correct to remove an outlier immediately from the dataset?
4. Why should preprocessing and transformation not be applied automatically? Give two examples in which a transformation changes the meaning of the original problem.
5. What is the difference between similarity and dissimilarity (distance)? Why is this distinction important when choosing a method for data analysis?
6. State the definition of a metric. Which properties must a distance measure satisfy in order to be a true metric?
7. Why is $1 -$ cosine similarity not always a true metric? Explain which metric property can fail and why this matters in practice.
8. Compare Euclidean distance, cosine similarity, and correlation with respect to scaling and translation. Which of them are invariant to scaling, and which are invariant to translation?
9. Consider two customers represented in different ways.

   (a) Customer A and customer B are described by their spending vectors across product categories, for example

   $$A = (120, 80, 0, 40, 10), \qquad B = (240, 160, 0, 80, 20),$$

where the components correspond to spending on groceries, clothing, electronics, books, and pharmacy products.

(b) The same two customers are also described by their total monthly spending over the last six months, in euros.

(c) Finally, the same two customers are described by standardized month-to-month spending profiles, where the goal is to compare the *pattern of ups and downs* rather than the absolute amounts.

For each of the three cases above, which proximity measure would be most appropriate, and why? Discuss whether Euclidean distance, cosine similarity, or correlation would be more meaningful in each situation.

10. Why can one-hot encoding followed by Euclidean distance be problematic for mixed-type data? What is the main idea of using a Gower-style distance instead?